Review: Circumference and area

1

a 50.3 cm b 113.1 cm c 40.8 cm

2

a 44.0 cm b 169.6 cm c 11.3 cm d 119.4 cm

3

a 2.9 m b 2.4 m

4

a 14.0 m b 22.3 m c 7.8 cm

5 7.00 cm

6 13.37 m

7 94.2 mm

8 25.1 m

9 100.5 cm

10 106.8 cm

11 311.0 cm

12 22.1 laps

13 345.6 m

14 163.4 m

15 8

16 39 788

17 231.0 km

18 $121\pi \text{ cm}^2$

19 113.1 cm^2

20 $25\pi \text{ cm}^2$

21
a 254.5 cm^2
b 452.4 cm^2

22 314.16 cm^2

23
a $r = 10.59$
b 66.5 cm

Additional questions

24
a $5\pi \text{ cm}$
b $8\pi \text{ ft}$
c $10\pi \text{ m}$
d $0.8\pi \text{ in}$

25
a $36\pi \text{ in}^2$
b $100\pi \text{ in}^2$
c $9\pi \text{ in}^2$
d $0.0625\pi \text{ in}^2$

26
a $A = 169\pi \text{ in}^2, C = 26\pi \text{ in}$
b $A = 2.25\pi \text{ yd}^2, C = 3\pi \text{ yd}$

27 $r = 6 \text{ ft}$

28 $196\pi \text{ in}^2$

29 About 53 in

30 The field will require 13 boxes of grass seed.

31 About 628 m

32

a $127\,270\,562.5625\pi \text{ cm}^2$

b $399\,629.57\,\text{cm}^2$

c Yes

33

a $126.5625\pi\,\text{in}^2$

b $120.3125\pi\,\text{in}^2$

c The probability of not hitting the bullseye is about 95%

34 Yes, based on Jameela's estimation, the diameter of the pond is about 32 ft, meaning the bridge is long enough to set across the center of the pond.

35 Larry is incorrect. The area of circle A is $A_A = 2.25\pi$ and the area of circle B is $A_B = 9\pi$. Since $9/2.25 = 4$ So the area of circle B is four times the area of circle A.

36 They should choose the 20-inch pizza. The two 14-inch pizzas has a total area of $98\pi \text{ in}^2$ square inches, whereas the single 20-inch pizza has a total area of $100\pi \text{ in}^2$.

37

a First you can measure the circumference of the tree. Then you can use the circumference to determine the radius of the tree. If divide the radius of the tree by the average width of a tree ring, the result will be the approximate age of the tree.

b The tree is about 9.5 years old.